

TBA4245 Geodesy. Lecture 18.8.2026.

Baseline transformation, part 1

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Topics:

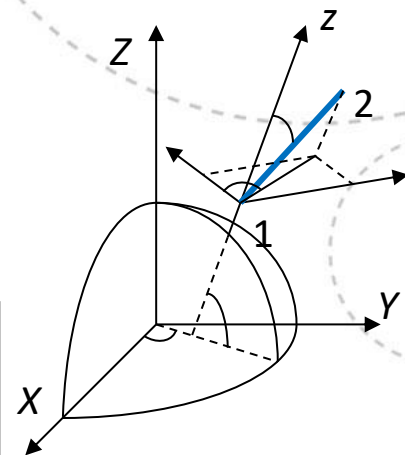
- Intro
- Baselines: From the ellipsoid to the (map) projection plane and NN2000
 - Baseline transformation, overview
 - Coordinate systems, Conversion, Transformation, SkTrans

Textbook:

Torge: Geodesy, Chapter 2.5, also 6.2 and 6,3 and pages 227-229

Alternative:

Skogseth/Holsen: "Basic GPS", NB: English version only, Chapters 10-11



Date	Lecture	Topic	Assignment. Readings
18.8.	2 hours	GNSS Baselines, part 1	Theory for Assignment no. 1
20.8.	2 hours	GNSS Baselines, part 1	Theory for Assignment no. 1

GNSS Baseline transformation from the ellipsoid to the (map) projection plane and NN2000

Before beginning the topic on baseline transformation, we will review the relevant information for this lecture.

Datum in classical and modern geodesy

	NGO1948	ED50	EUREF89	WGS84
Area covered:	Norway	Europe	Europe	Global
Ellipsoid:	Modified Bessel	International	GRS1980	WGS84
a (semimajor axis):	6377492,0176	6378388,0000	6378137,0000	6378137,0000
b (semiminor axis):	6356173,5083	6356911,9461	6356752,31414	6356752,3142
f (flattening):	1/299,15281285	1 / 297,000	1/298,257223563	1/298,257222100
Geoid-ellipsoidal separation in Norway (geoidehøyde):	Ca. 0 – 12 m	Ca. -2 - 14 m	Ca. 17 - 48 m	Ca. 17 - 48 m
Map projection:	Gauss-Krüger	UTM	UTM	UTM

Classical Geodesy

Modern Geodesy

EUREF89 and NN2000 in Norway

EUREF89 is the official geodetic reference frame in Norway

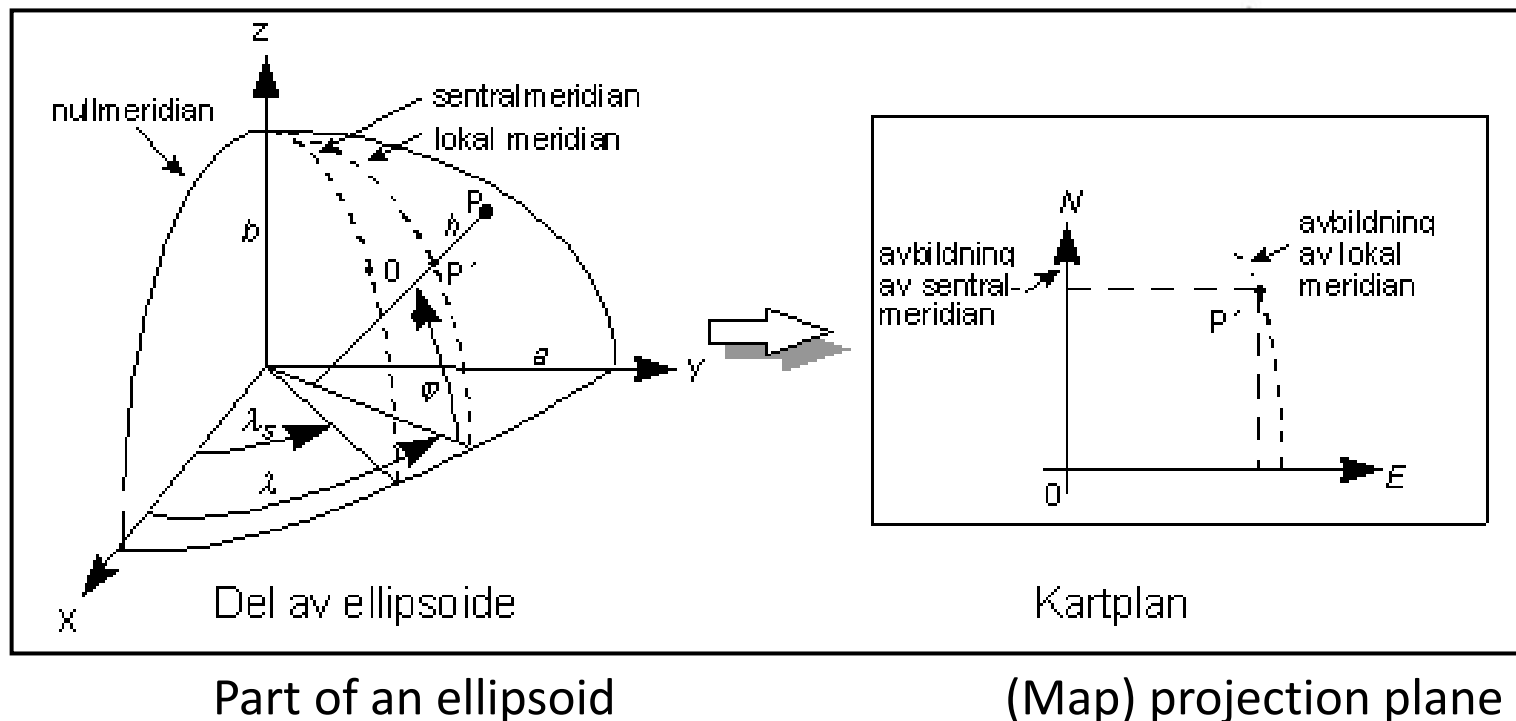
NN2000 is the national vertical (height) system in Norway

<http://etrs89.ensg.ign.fr/>

<https://www.kartverket.no/NN2000>

<https://www.kartverket.no/Posisjonstjenester/bruke-referanserammer/Referanserammer-for-Noreg/>

From/to the ellipsoid to/from the (map) projection plane and (for heights) to the national vertical system



Figures from
Norwegian
mapping
authority
(Kartverket)

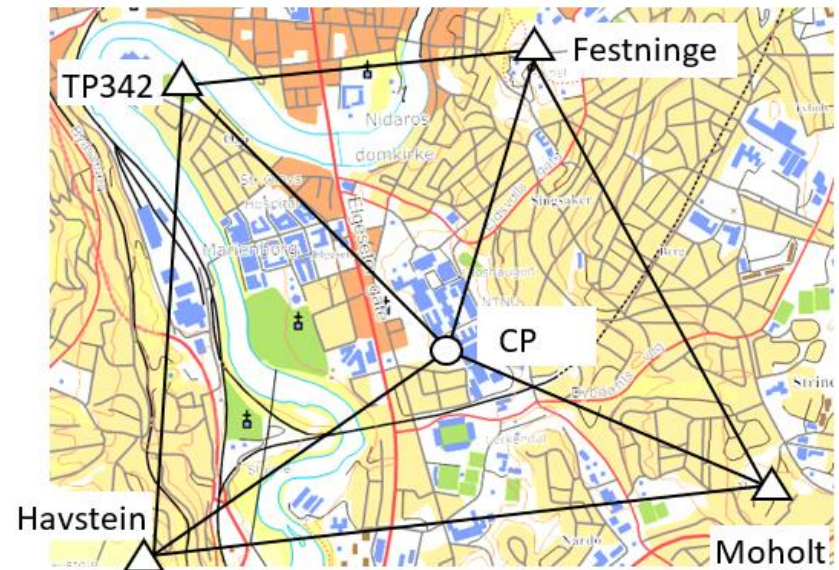
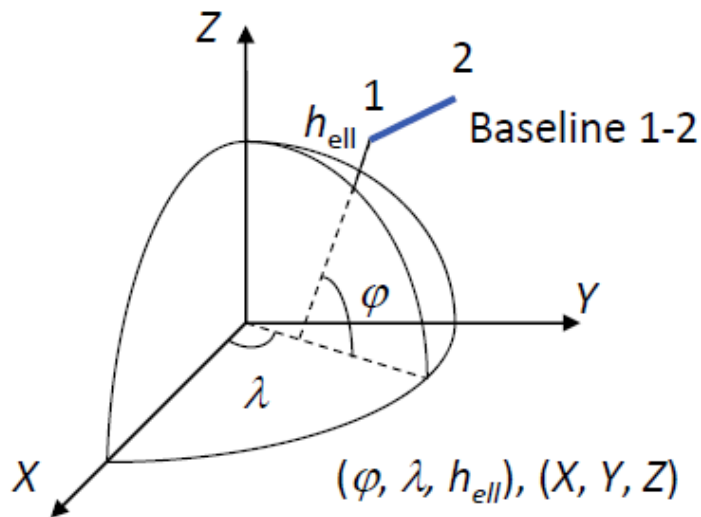
Horizontal coordinates

Heights

Two methods for obtaining horizontal coordinates in the national frame EUREF89 UTM and heights in the national system NN2000 in Norway

(1) The method used in the course Theoretical geomatics to transform GNSS baselines to coordinates in the (map) projection plane and heights referred to the NN2000:

- baselines ($\Delta X, \Delta Y, \Delta Z$) are **processed** in a Cartesian 3D XYZ coordinate system.
- Coordinates of the unknown points are **estimated with Least Squares method in Cartesian coordinate system**.
- Then **a transformation** of the XYZ values to N(orth), E(ast) in the (map) projection plane and H(eights) in the national vertical system NN2000. We used the transformation program SkTrans (or GISLineTrans)

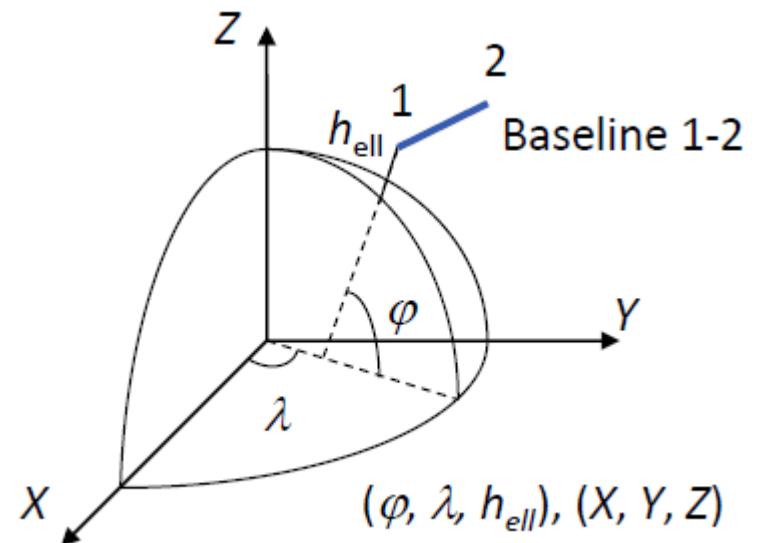


Two methods for obtaining horizontal coordinates in the national frame EUREF89 UTM and heights in the national system NN2000 in Norway

(2) Another method in Norway: from GNSS baselines to coordinates in the (map) projection plane and heights referred to the NN2000:

- Baselines are **processed** in a Cartesian 3D XYZ coordinate system.
- **Transform** the GNSS baselines ($\Delta X, \Delta Y, \Delta Z$) to observations in the (map) projection plane and national height system: Distance, bearing, height difference (or zenith angle)
- Then the coordinates and heights are **estimated with least squares method in the (map) projection plane and NN2000 height system** (EUREF89-UTM-zone 32 and NN2000).

$$\begin{bmatrix} \Delta X \\ \Delta Y \\ \Delta Z \end{bmatrix} = \begin{bmatrix} X_{Unknown} \\ Y_{Unknown} \\ Z_{Unknown} \end{bmatrix} - \begin{bmatrix} X_{Known} \\ Y_{Known} \\ Z_{Known} \end{bmatrix}$$



Two methods for obtaining horizontal coordinates in the national frame EUREF89 UTM and heights in the national system NN2000 in Norway

Before starting the second method, we will review some topics from previous courses

GNSS baselines

Results of a GNSS baseline computation:

ΔX , ΔY , ΔZ :

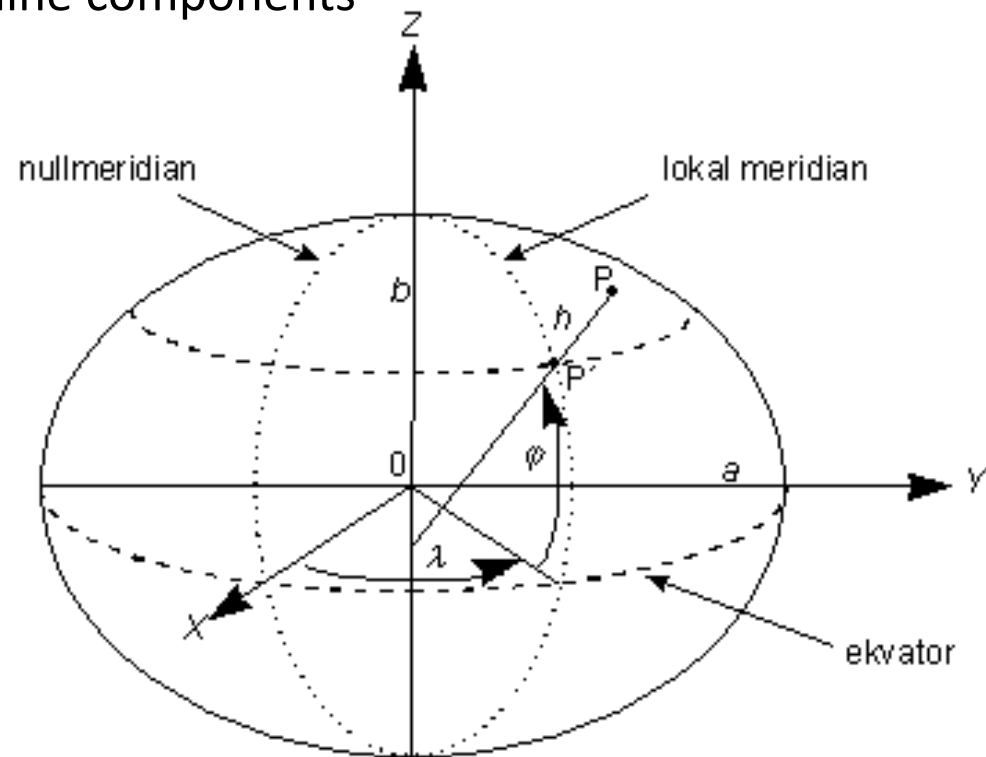
Cartesian coordinate differences are expressed in a “Conventional Global Geocentric Coordinate System”, also referred to as a “Conventional Terrestrial Reference System (CTRS)” (sometimes called CTS). The same system is also known as the “Global Geocentric Ellipsoidal System” or “Earth-Centered, Earth-Fixed (ECEF)” system. In our case, this system is WGS84. WGS 84 is an “Earth-Centered, Earth-Fixed (ECEF)” system and is closely aligned with the “International Terrestrial Reference System (ITRS)”.

Observations in WGS84.

Coordinates in EUREF89 and heights in NN2000 in Norway.

From the baseline computation we obtain:

- Geocentric coordinate differences, ΔX , ΔY , ΔZ
- Geodetic (ellipsoidal) coordinates and ellipsoidal heights of the start and the end of the baseline.
- The standard deviations σ_x , σ_y , σ_z of respectively ΔX , ΔY , ΔZ
- The correlation matrix of the baseline components



"Nullmeridian": Zero meridian

Absolute coordinates have a lower accuracy than the coordinate differences. Why?

An example of a GNSS baseline calculation

Result of a baseline calculation:

- The coordinates of an (rover) unknown point calculated from given coordinates of a reference point
- The coordinate differences ΔX , ΔY and ΔZ in the WGS84, Cartesian coordinates.
- The standard deviations (s) of the ΔX , ΔY and ΔZ . See that the value of the standard deviations calculated together with the baselines: They are too low.
- Slope distance.

Rov:st2 Ref:st46 Amb: Y Proc: L1+L2 phase 21.03.00 12:54:30									
Cartesian :									
X	2814807.7505	m	Y	516795.6314	m	Z	5681070.0818	m	
ΔX	1658.3013	m	ΔY	-1261.6960	m	ΔZ	-726.3491	m	
sX	0.0001	m	sY	0.0001	m	sZ	0.0002	m	
Geodetic :									
Lat	63 25 2.99595	N	Lon	10 24 12.88320	E	h	129.8096	m	
Δ Lat	-51.01155		Δ Lon	-1 51.07770		Δ h	-22.0354	m	
sLat	0.0001	m	sLon	0.0001	m	sh	0.0002	m	
Distance :									
Slope	2206.6769	m							
sSlope	0.0001	m							

1/3 printouts from post processing: (Baselines st46 to s2)

Another version of processing results compared to the one shown on the previous page

Reference Id,st46
 Rover Id,s2
 Stored Status,Yes
 Ambiguity Status,yes
 Start,03/21/2000 12:54:30
 End,03/21/2000 17:53:50
 Duration,4h 59' 20''
 Type,Static
 Solution type,Phase
 Frequency,L1 + L2
dx, 1658.3100
dy,-1261.7004
dZ,-726.3501
 Posn. Qlty, 0.0001
 Hgt. Qlty, 0.0002
 Posn. + Hgt. Qlty, 0.0002
 Slope Distance, 2206.6863
 Sd. Slope Distance, 0.0001

Ref. Ant. Hgt., 0.1370
 Rov. Ant. Hgt., 0.3550
 Sd. X, 0.0001
 Sd. Y, 0.0001
 Sd. Z, 0.0002
M0, 0.3828
Q11,0.00000006
Q12,0.00000001
Q13,0.00000005
Q22,0.00000003
Q23,0.00000001
Q33,0.00000023
 GDOP (min),1.9
 PDOP (min),1.7
 HDOP (min),0.7
 VDOP (min),1.5

GDOP (max),27.2
 PDOP (max),22.2
 HDOP (max),6.6
 VDOP (max),21

13 2/3 printouts from post processing

		Reference: st46	Rover: s2
Coordinates:			
	X:	2813149.4500 m	2814807.7600 m
	Y:	518057.3270 m	516795.6266 m
	Z:	5681796.4300 m	5681070.0799 m
Solution type:		Phase	
Frequency:		L1 and L2	
Ambiguity:		Yes	

Quality:	Sd. X: 0.0001 m	Sd. Y: 0.0001 m	Sd. Z: 0.0002 m
	Posn. Qlty: 0.0001 m	Hgt. Qlty: 0.0002 m	Sd. Slope: 0.0001 m
M0:	0.3828 m		
Cofactor matrix Qxx:	0.00000006	0.00000001	0.00000005
		0.00000003	0.00000001
			0.00000023
Baseline vector:	dX: 1658.3100 m	dY: -1261.7004 m	dZ: -726.3501 m
	Slope: 2206.6863 m	dHgt: -22.0329 m	
DOPs (min-max):	GDOP: 1.9 - 27.2		
	PDOP: 1.7 - 22.2	HDOP: 0.7 - 6.6	VDOP: 1.5 - 21.2

3/3 Ascii-format (export/import):

```

@%Unit:                m
@%Coordinate type:      Cartesian
@%Reference ellipsoid:  WGS 1984
@#s2                    2814807.7600    516795.6266    5681070.0799    MEAS
12
@#s2                    2814787.6856    516811.3629    5681094.0626    NAV
12
@#st46                  2813149.4500    518057.3270    5681796.4300    REF
12
@+st46                  2813149.4500    518057.3270    5681796.4300
@-s2                    1658.3100    -1261.7004    -726.3501
@=      0.3828    0.00000006    0.00000001    0.00000005    0.00000003    0.00000001
0.00000023
@:      0.1370    0.0000
@;      0.3550    0.0000
@*21.03.2000 12:54:30
@E      0.0000    0.0000    0.0000    0.0000
@#st46    2813149.4500    518057.3270    5681796.4300

```

The variance-covariance-matrix of the baseline

An example

The calculated values of the standard deviation of the unit weight is $s_0 = 0,3957$.

The *co-factor matrix*, $\mathbf{Q}_{xyz}$, unit m^2

The matrix is full and symmetric (the upper triangle is shown in the table). The elements of the matrix show that *the three coordinate differences are correlated*. Why is this the case?

a posteriori rms : 0.3957			
Co-factor matrix (upper triangle [m*m]) :			
	qx	qy	qz
qx	+5.9344000E-008	+5.1850000E-009	+5.4495001E-008
qy		+2.5848999E-008	+9.3720001E-009
qz			+2.2508399E-007

The standard deviation of ΔX , ΔY and ΔZ in the table can be calculated from the values of the standard deviation of the unit weight and the co-factor matrix

The variance-covariance-matrix: $\mathbf{C}_{XYZ} = s_0^2 \cdot \mathbf{Q}_{XYZ}$, unit mm^2

$$\mathbf{C}_{XYZ} = \begin{pmatrix} \sigma_{XX}^2 & \sigma_{XY} & \sigma_{XZ} \\ \sigma_{XY} & \sigma_{YY}^2 & \sigma_{YZ} \\ \sigma_{XZ} & \sigma_{YZ} & \sigma_{ZZ}^2 \end{pmatrix} = \begin{pmatrix} 0,00929 & 0,00081 & 0,00853 \\ 0,00081 & 0,00404 & 0,00147 \\ 0,00853 & 0,00147 & 0,03524 \end{pmatrix}$$

Convert versus transform

Unique mathematical operation = **convert** →

Frame 1

Cartesian:

X,Y,Z



Geodetic

φ, λ, h



Map projection

x,y or *N,E* and H

7-parameter

transformation

Frame 2

Cartesian:

X,Y,Z



Geodetic

φ, λ, h



Map projection

x,y or *N,E* and H

4-parameter

transformation

← Empirical operation = **Transformation** →

Based on parameter estimation in «**common points**»

$$X = (N + h) \cos \varphi \cos \lambda,$$

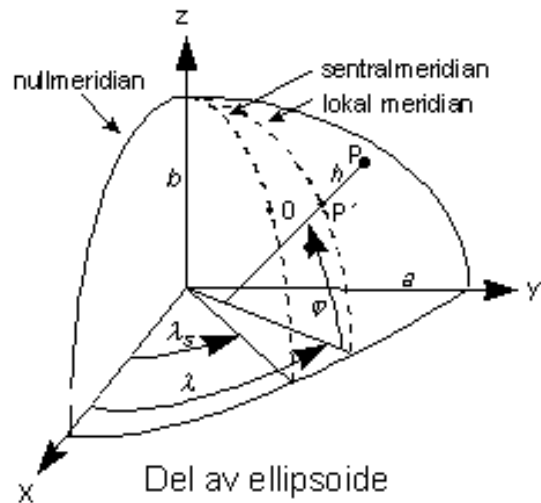
$$Y = (N + h) \cos \varphi \sin \lambda,$$

$$Z = \left(\frac{b^2}{a^2} N + h \right) \sin \varphi,$$

Convert Ellipsoidal to
Cartesian coordinates

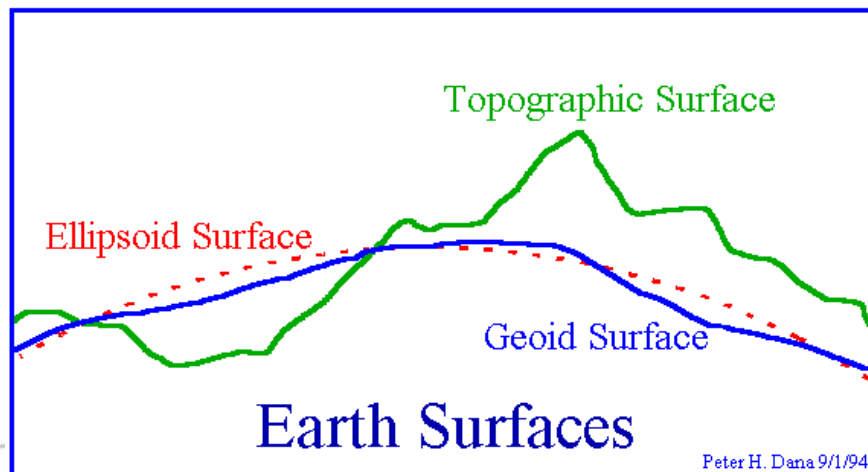
$$N = \frac{a^2}{\sqrt{a^2 \cos^2 \varphi + b^2 \sin^2 \varphi}}$$

Conversion to/from geodetic coordinates (latitude and longitude) and ellipsoidal heights on the WGS84 ellipsoid



- Geocentric Cartesian coordinates: X, Y, Z
- Longitude, latitude, ell. height: λ, φ, h
- Coordinates in a (map) projection plane and height: N, E, H

Conversion gives unambiguous results.
Formulas found in Hofmann-Wellenhof "GNSS".



- Ellipsoidal heights
- Orthometric heights (old NN1954)
- **Normal heights (NN2000)**
- Geoid undulation (geoid-ellipsoid separation, *geoidehøyde*)

Transformation program: SkTrans

An official transformation program for Norwegian territory of the reference frames NGO1948, ED50 and EUREF89. In this example the from system is EUREF89-UTM-zone 32 with the Northing and Easting coordinate values, and the height is ellipsoidal height of 106.047 m. The to system is EUREF89 Latitude / Longitude (Bredde/Lengde), and height in the height system NN2000. The transformed values are shown in the white box to the right

Enkeltpunkt

Fra system

Euref89 UTM 32

Koordinat for transformasjon

N 7033994.270 E 570391.267

Høyde

Ellipsoidisk 106.047 m

Til system

Euref89 Bredde/Lengde

Høyde

NN2000

Resultat

N 63.427480898°
E 10.410436463°

N 63° 25.6488539'
E 10° 24.6261878'

N 63° 25' 38.93123"
E 10° 24' 37.57127"

Høyde: 66.682

Buttons: Fritt format, Sosi, Innstillinger, Transformer, Avslutt, Transinfo

GISLINE Trans

Program for converting within, or transformation between reference frames.

The screenshot displays the GISLINE Trans application window. The title bar reads "GISLINE Trans". The menu bar includes "Transformer", "Legg til", "Ta ut", "Åpne base", "Om ...", and "Hjelp". The main interface is divided into two primary sections: "Fra" (From) and "Til" (To).

Fra (From) coordinates:

- X: 2815415.137
- Y: 518305.734
- Z: 5680680.336

Til (To) coordinates:

- X: 7031952.892
- Y: 571469.041
- H: 170.838

Below the coordinate fields is a tabbed interface with the following tabs: "Enkeltpunkt", "Quadri", "Fritt format", "SOSI", "Land", "DXF", "TerraScan", "KOF", "VGR", and "LAS". The "Enkeltpunkt" tab is currently selected.

Under the "Enkeltpunkt" tab, there are two columns of settings:

FRA (Left Column):

- Datum: EUREF89
- Koordinatsystem: Kartesiske koordinater
- Akse / sone: (empty)
- Enhet: Meter
- Vertikal datum: Ortometriske høyder

TIL (Right Column):

- Datum: EUREF89
- Koordinatsystem: Kartplan
- Akse / sone: UTM-Sone 32
- Enhet: Meter
- Vertikal datum: Ellipsoidiske høyder

At the bottom of the main area, there are two fields:

- Type: skt2nor1.dll
- Geoide: href2008a.bin

The bottom status bar shows "Ready" on the left and "NUM" on the right. A small tabbed interface at the very bottom includes "SK-trans", "GL-trans", "Egendefinert", "H-trans", and "Parametre".

We begin with the second method to transfer GNSS baselines to the (map) projection plane and the NN2000 height system.

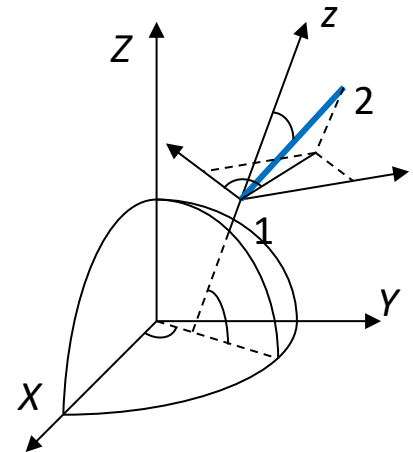
GNSS baselines to the (map) projection plane and the NN2000 height datum

The second method for obtaining coordinates and heights in the (map) projection plane and the NN2000 height system from the measured GNSS baselines is:

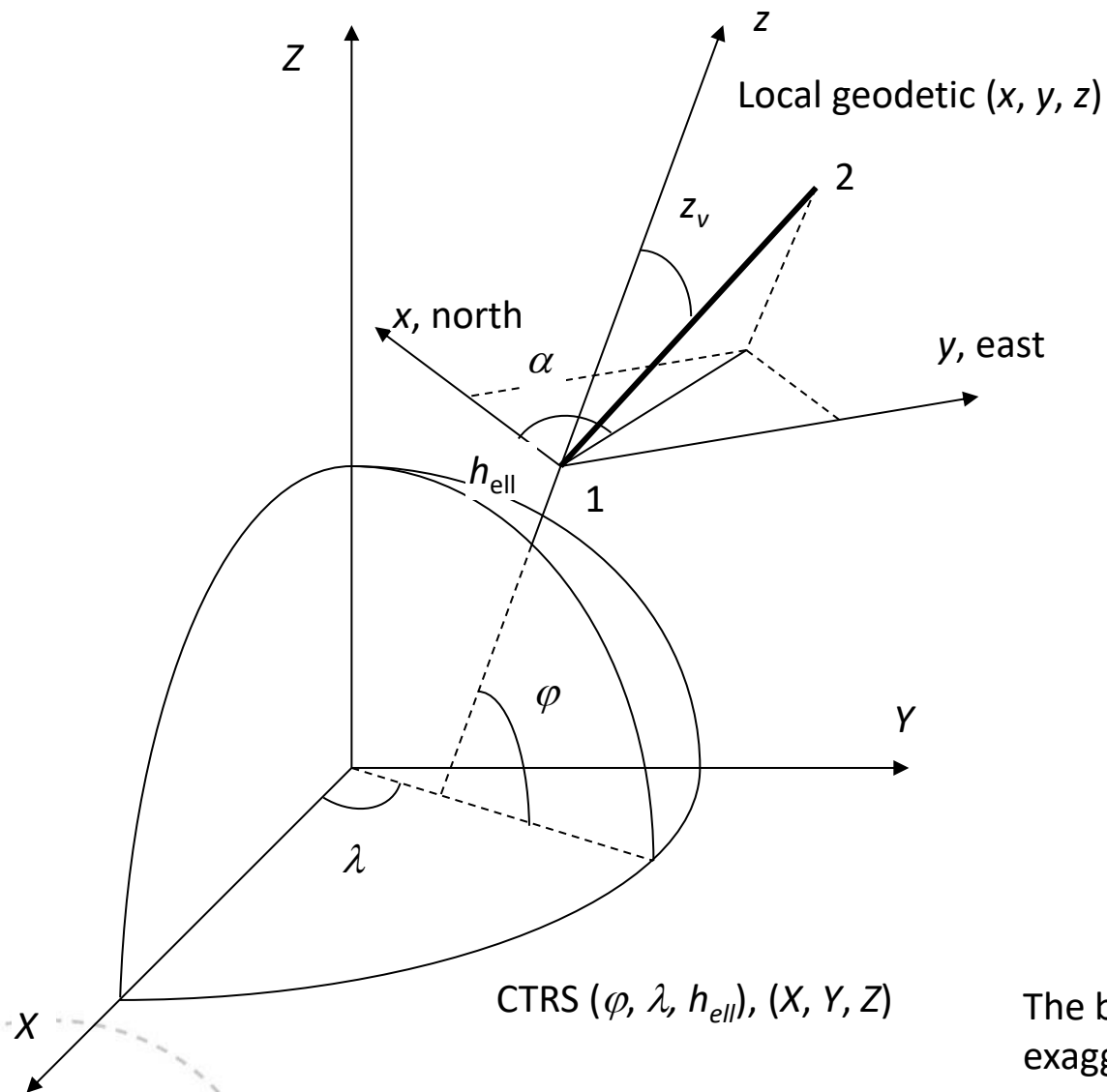
1. Transform the GNSS baselines to observables in the (map) projection plane and height system (**distance, bearing, zenith angle/height difference**)
2. Calculate the coordinates and heights in the (map) projection plane and height system **using the Least Squares Method (LSM)** (“Minste Kvadraters metode”)

Two main steps for the transformation in 1:

- 1.1 From a Global geocentric **Cartesian coordinate system** to a **local geodetic coordinate system**
- 1.2 From the local geodetic coordinate system to the **(map) projection plane (for instance UTM) and the NN2000 height system**



1. The components of the baseline, ΔX , ΔY , ΔZ , converted to a topocentric system



Topocentric coordinate system (local horizontal system)

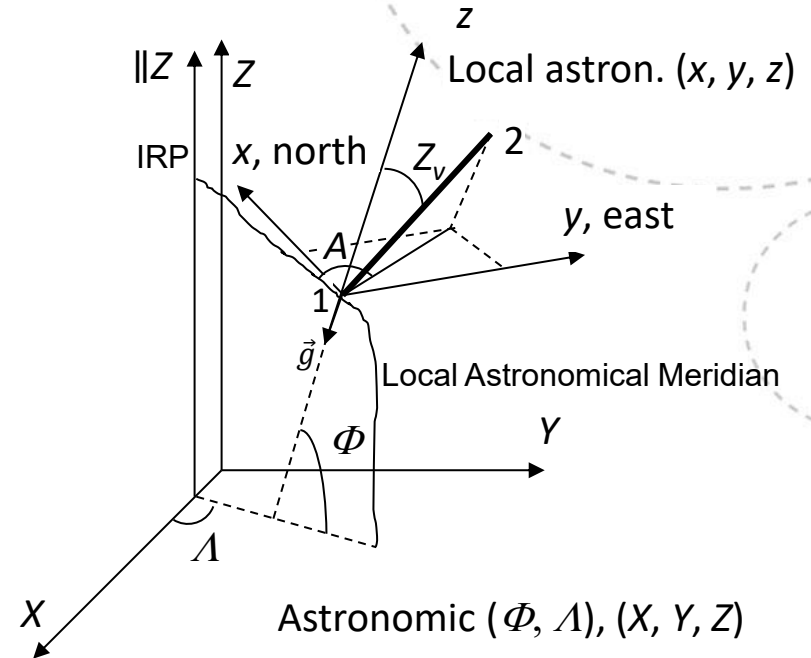
1: The Local **Astronomical** coordinate system (LA)
2: The Local **Geodetic** coordinate system (LG)

The coordinate systems are called *topocentric* (they have the origin at the site of the observer on the surface of the Earth)

The baseline 1-2 is somewhat exaggerated (!) in length on the figure...

The local astronomical (LA) coordinate system is linked to the geoid and the plumb line

- Origin (at the site of the observer on the surface of the Earth)
- The z axis (the gravity vector defines the negative z axis)
- The x axis is defined as the line of intersection between the horizontal plane at the observation point and the vertical plane at that point that contains the special north pole known as the International Reference Pole (IRP). The position of the IRP is determined by international agreement and thus defines a point on the Earth's rotation axis. The direction of the x axis gives *the astronomic north*. Note that we use a parallel to the rotation axis in LA system.
- The y axis is pointing towards east. The coordinate system is therefore a left-hand system.



Note that the observations are made in the LA system

The local geodetic (LG) coordinate system is linked to the ellipsoid and the ellipsoid normal

(a) Origin

One end of the baseline, here point 1. (at the site of the observer on the surface of the Earth)

(b) The z axis

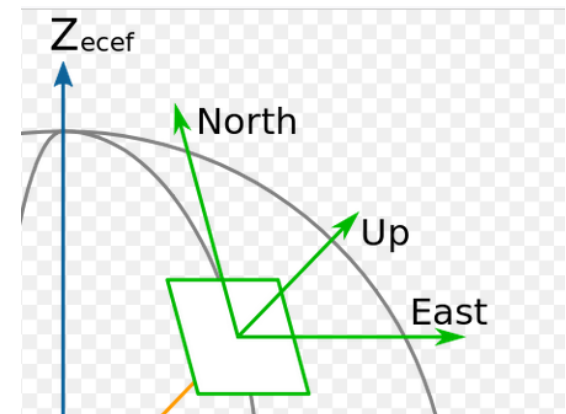
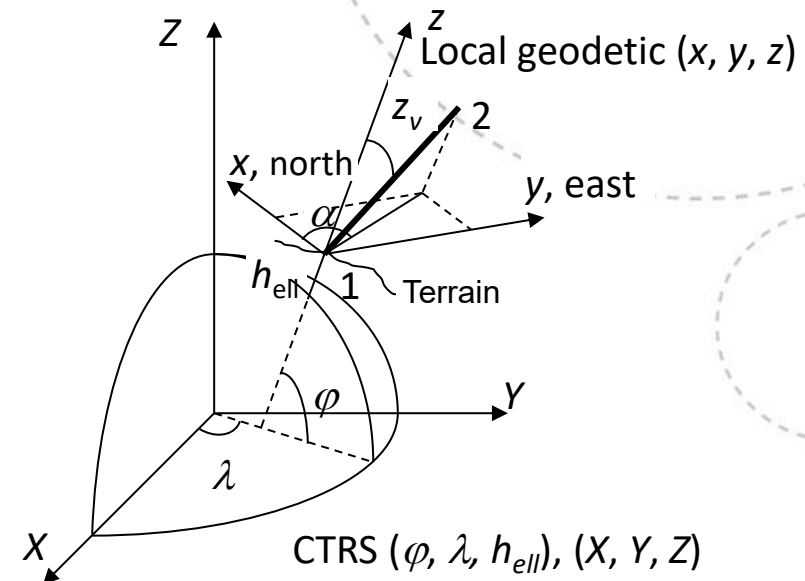
Outward Ellipsoid normal through point 1

(c) The x axis

In the tangent plane (the plane normal to the z axis in point 1). Towards the rotation axis, Z , *geodetic north*. It lies in the geodetic meridian plane defined by the ellipsoid normal at point 1 and the Z (semi-minor axis) of the reference ellipsoid.

(d) The y axis is pointing towards east.

The coordinate system is therefore a left-hand system.



The differences between LA and LG systems (mathematical relations) are given by 3 formulas.

The **deflection of the vertical**, also called the deflection of the plumb line (= *loddavviket*). The deflection has 2 components, ξ and η (ksi and eta):

1) **The northern component:** $\xi = \Phi - \varphi$

2) **The eastern component:** $\eta = (\Lambda - \lambda) \cdot \cos \Phi$

The difference in astronomic azimuth A and geodetic azimuth α is the 3rd formula, the **Laplace equation**:

3) **Laplace:** $A - \alpha = (\Lambda - \lambda) \cdot \sin \Phi = \eta \cdot \tan \Phi$

The symbols used in the formulas are:

Λ : astronomical longitude

Φ : astronomical latitude

λ : geodetic longitude

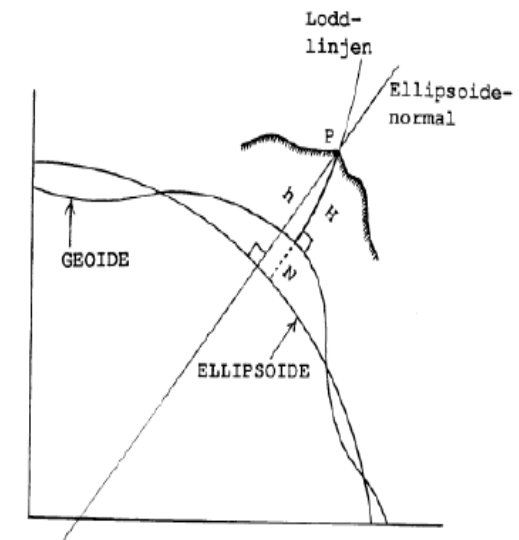
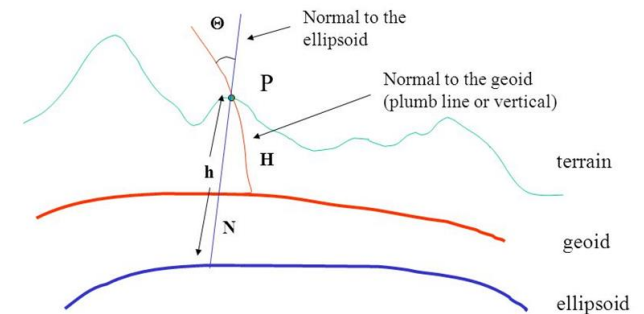
φ : geodetic latitude

Θ - angle between the normal to the ellipsoid and the vertical direction (normal to the geoid), so-called deflection of the vertical

H - orthometric height

h - ellipsoidal height $h = H + N$

N - geoid undulation (computed from geoid model provided by NGS)



Loddlinjen = The plumb line or the vertical

Transformation between the LG and LA system

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix}^{LG} = R_3(\Delta A) R_2(-\xi) R_1(\eta) \begin{bmatrix} x \\ y \\ z \end{bmatrix}^{LA}$$

$$\Delta A = A - \alpha = (\Lambda - \lambda) \sin \varphi = \eta \tan \varphi$$

$$\xi = \Phi - \varphi \quad \eta = (\Lambda - \lambda) \cos \varphi \quad \begin{matrix} z = Z_V \\ z' = Z_V \end{matrix}$$

we can write:

$$\begin{bmatrix} \cos \alpha \sin z' \\ \sin \alpha \sin z' \\ \cos z' \end{bmatrix} = \begin{bmatrix} 1 & \Delta A & \xi \\ -\Delta A & 1 & \eta \\ -\xi & -\eta & 1 \end{bmatrix} \begin{bmatrix} \cos A \sin z \\ \sin A \sin z \\ \cos z \end{bmatrix}$$

using a Taylor series for the left hand side around (A, z) , we get:

$$\begin{aligned} (A - \alpha) \begin{bmatrix} \sin A \sin z \\ -\cos A \sin z \\ 0 \end{bmatrix} + (z - z') \begin{bmatrix} -\cos A \cos z \\ -\sin A \cos z \\ \sin z \end{bmatrix} &= \\ &= \begin{bmatrix} 0 & \Delta A & \xi \\ -\Delta A & 0 & \eta \\ -\xi & -\eta & 0 \end{bmatrix} \begin{bmatrix} \cos A \sin z \\ \sin A \sin z \\ \cos z \end{bmatrix} \end{aligned}$$

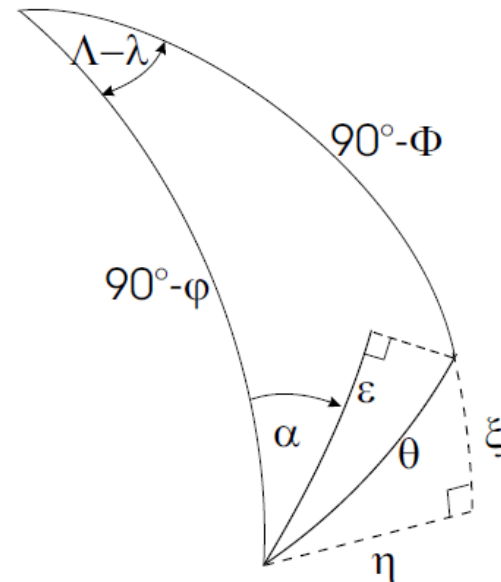
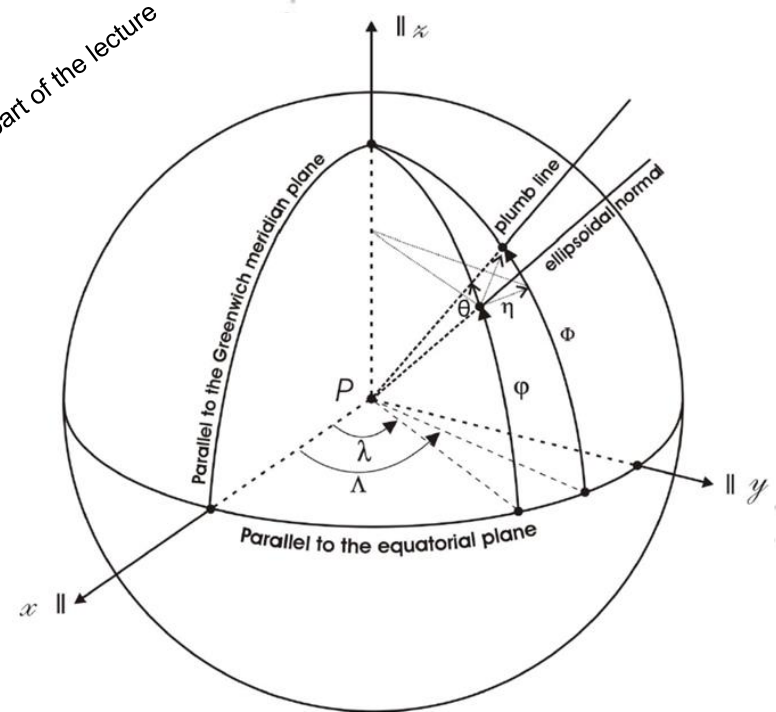
The third equation gives:

$$z - z' = -\xi \cos A - \eta \sin A$$

multiplication of the first equation by $\sin A$, the second by $\cos A$, subtraction of the second from the first, and substitution for ΔA , we get

$$A - (\xi \sin A - \eta \cos A) \cos z - \alpha = \eta \tan \varphi$$

Not part of the lecture



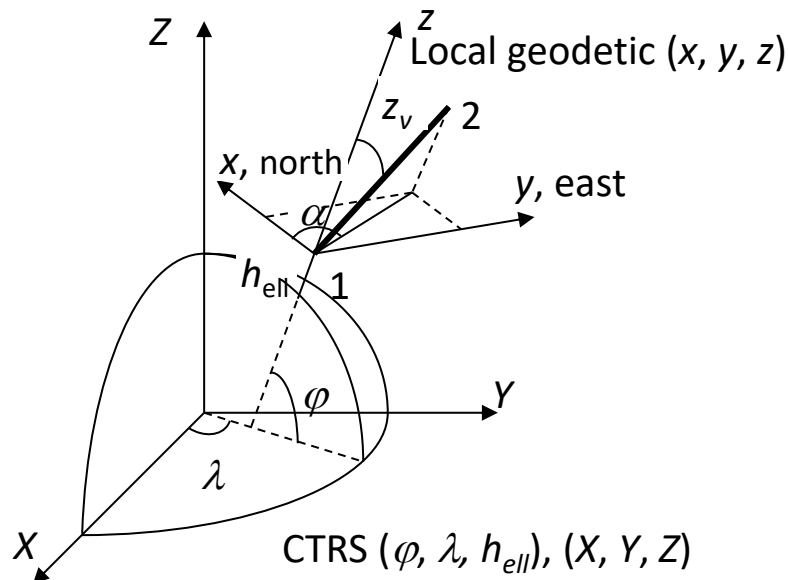
From: The global coordinate system:

$\Delta X, \Delta Y, \Delta Z$

Cartesian coordinates, CTRS=WGS84=ECEF

To: Local geodetic coordinate system:

x, y, z



From:
$$\begin{pmatrix} \Delta X \\ \Delta Y \\ \Delta Z \end{pmatrix}_{CTRS}$$

to:
$$\begin{pmatrix} x \\ y \\ z \end{pmatrix}_{LG}$$

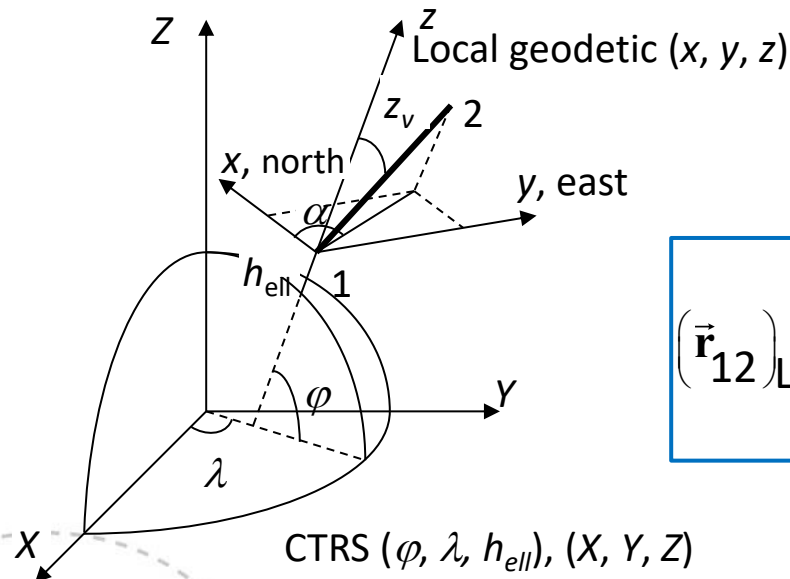
$$\begin{pmatrix} x \\ y \\ z \end{pmatrix}_{LG}$$
 are also named
$$\begin{pmatrix} \text{North} \\ \text{East} \\ \text{Up} \end{pmatrix}_{LG}$$

From geocentric CTRS to local geodetic system (LG)

Geographic coordinates (φ and λ) are given for point 1, and they are referred to the CTRS=WGS84 ellipsoid.

Conversion from CTRS to LG of a baseline between point 1 and point 2, where the origin of the LG system is at point 1:

$$\begin{pmatrix} \vec{r}_{12} \end{pmatrix}_{\text{LG}} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}_{\text{LG}} = \mathbf{P}_2 \cdot \mathbf{R}_2(\varphi - 90^\circ) \cdot \mathbf{R}_3(\lambda - 180^\circ) \cdot \begin{bmatrix} \Delta X \\ \Delta Y \\ \Delta Z \end{bmatrix}_{\text{CTRS}}$$



$$\begin{pmatrix} \vec{r}_{12} \end{pmatrix}_{\text{LG}} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}_{\text{LG}} = \begin{bmatrix} -\sin\varphi\cos\lambda & -\sin\varphi\sin\lambda & \cos\varphi \\ -\sin\lambda & \cos\lambda & 0 \\ \cos\varphi\cos\lambda & \cos\varphi\sin\lambda & \sin\varphi \end{bmatrix} \cdot \begin{bmatrix} \Delta X \\ \Delta Y \\ \Delta Z \end{bmatrix}_{\text{CTRS}}$$